Exaggeration Detection in Health Science Press Releases

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1 Abstract

Public trust in scientific findings is essential for translating research into real-world applications. This trust can be undermined when news reports of scientific discoveries misrepresent the findings of the underlying research. To address this issue, we develop a model to detect exaggeration and understatement of scientific findings in news communications intended for the general public. We leverage a curated dataset consisting of press releases paired with the corresponding scientific paper abstracts, which have been manually annotated by experts to indicate whether the press release accurately reflects, exaggerates, or downplays the findings of the original manuscript. Given the highly technical nature of scientific abstracts, we compare the performance of PubMedBERT and RoBERTa, two encoder models with similar architectures but different training corpora, to determine whether pretraining on scientific literature rather than general English text improves exaggeration detection in this domain. Because the dataset is relatively small, we apply Low-Rank Adaptation (LoRA) to efficiently fine-tune the models and maximize performance under limited training data. Finally, we compare the performance of PubMedBERT and RoBERTa with DeBERTa and ChatGPT to assess whether architectural improvements outweigh the benefits of domain-specific pretraining. We demonstrate that PubMedBERT outperforms RoBERTa, DeBERTa, and ChatGPT in classifying press release exaggeration and that LoRA further improves the ability to detect exaggeration.

2 Introduction

With accelerating news cycles and intense competition for public attention, news organizations are increasingly incentivized to produce bold, attention-grabbing headlines. While such strategies can increase readership, they can also distort the communication of scientific findings, leading to exaggerated claims, public misinformation, and the potential erosion of public trust in research. Surveys suggest that public confidence in scientific institutions has decreased in recent years, with only 53% of Americans reporting that they believe that scientists provide unbiased conclusions about their research [9]. These challenges highlight the importance of developing tools that can identify when scientific findings are inaccurately represented in communications for the general public. In this paper, we explore whether natural language processing and domain-specific language models can help identify mismatches between what scientific papers claim and the information that is released to the public.

Prior work has investigated discrepancies between scientific publications and their public dissemination. Early studies relied on manual analysis of university press releases and the corresponding peer-reviewed articles to identify exaggeration or misrepresentation of research findings [12, 2]. Because large-scale manual annotation is costly and difficult to scale, subsequent research applied machine learning and natural language processing techniques to classify the strength of claims made in scientific abstracts and press releases. More recently, models have been trained on paired abstracts and press releases, allowing them to learn the relationship between scientific findings and public statements more directly [13]. Together, these studies represent important steps toward the automated detection of exaggerated or minimized scientific claims in public reporting.

However, most previous approaches rely on language models pretrained on general-domain text. Scientific writing often contains highly specialized terminology and domain-specific conventions, suggesting that models trained on scientific corpora may be better able to identify subtle mismatches between research findings and public communications. In this work, we investigate whether domain-specific pretraining provides greater benefit for exaggeration detection than improvements in transformer architecture. Specifically, we compare the performance of RoBERTa and PubMedBERT, two transformer encoders with similar architectures but different pretraining corpora. Because the available dataset of paired abstracts and press releases is relatively small, we employ Low-Rank Adaptation (LoRA) to enable efficient fine-tuning of these models. We then compare their performance to a more advanced encoder architecture (DeBERTa) and a large language model. Finally, to evaluate the feasibility of deploying such models in resource-constrained settings, we examine whether model performance is preserved under post-training quantization.

Our contributions are fourfold:

1. We evaluate whether domain-specific language model pretraining improves the detection of exaggerated or minimized scientific claims in press releases when compared to models trained on general-domain text. Formally, our hypothesis is:

$$H_0: \ \mu_{\text{PubMedBERT}} \leq \mu_{\text{RoBERTa}}$$

$$H_A: \ \mu_{\text{PubMedBERT}} > \mu_{\text{RoBERTa}}$$

2. We apply parameter-efficient fine-tuning using LoRA to enable effective training on a limited dataset of paired scientific abstracts and press releases, and compare its performance to full-parameter fine-tuning.

3. We compare the domain-specific PubMedBERT model to more modern architectures, including DeBERTa-v3 and a large language model (ChatGPT 5.4), to assess whether architectural advances can offset the benefits of domain-specific pretraining.

4. We assess whether model performance remains robust under post-training quantization, providing insight into the feasibility of deploying such models in limited-resource environments.

We hypothesize that domain-specific pretraining will yield improved macro F1 scores compared to general-domain pretraining, even when using more advanced model architectures, and that performance will be further improved through LoRA-based fine-tuning. We additionally hypothesize that performance will persist under post-training quantization (PTQ). Our results support these hypotheses, with the highest performance achieved by applying LoRA to PubMedBERT.

3 Background

Often, peer-reviewed scientific papers are published in journals requiring expensive subscriptions, restricting their availability to individuals with affiliations with large research institutions or universities. To allow their findings to be available more widely, research organizations frequently generate press releases, which then inform news articles for the general public. At each stage of this chain, distortions can occur from the original scientific findings, resulting in public misinformation, confusion, and skepticism. Manual coding of scientific abstracts and their corresponding university and news press releases found that 33% contained exaggerated causal claims, and 36% included exaggerated conclusions about human outcomes based on animal studies. Of the press releases that contained exaggeration, 81%, and 86% of corresponding news stories, respectively, contained similar exaggeration, findings that were replicated 5 years later [2, 12].

With thousands of scientific articles published each day, manual annotation is not a scalable approach for detecting exaggeration in press releases and news articles. To address this limitation, prior work has applied NLP techniques to identify exaggerated claims more efficiently. Early efforts utilized classification models such as multinomial Naive Bayes and support vector machines to predict claim strength in scientific articles and press releases [7], while later work incorporated transformer-based models to classify causal claims in observational research, a common form of exaggeration [14, 15]. Patro et al. (2021) propose a

pipeline that extracts claim-related phrases using BERT-based sequence labeling and subsequently applies traditional classifiers such as Naive Bayes, Random Forest, SGD, and XGBoost to predict claim strength, demonstrating improved performance over earlier baselines such as Li (2017) [10, 7]. While these approaches predict claim strength separately for scientific papers and press releases and compare the outputs to detect exaggeration [15, 10], Wright and Augenstein (2021) instead model paired inputs directly at inference time by reformulating exaggeration detection as a cloze-style prompting task using pretrained language models [13].

Building on this prior work, we investigate whether domain-specific pretraining provides greater benefit for exaggeration detection than improvements in transformer architecture by directly modeling paired scientific abstracts and press releases with pretrained encoder models. We further investigate whether model performance is preserved under post-training quantization (FP16 and INT8), to determine if deploying such models in a resource constrained environment is feasible. Our hypothesis is motivated by prior findings that domain-specific pretrained models such as BioBERT improve claim strength classification in science-related texts [15]. However, to our knowledge, our study is the first to evaluate domain-specific pretraining in a setting that directly models paired scientific abstracts and press releases for exaggeration detection.

4 Methods

4.1 Dataset

We train and evaluate our models on 663 paired scientific abstracts and press releases compiled by Wright and Augenstein (2021) from expert-annotated datasets introduced by Sumner et al. (2014) and Bratton et al. (2019), and made publicly available via Hugging Face [13, 12, 2]. Each pair is labeled as "same," "exaggerates," or "downplays," indicating the manually annotated relationship between the abstract and press release. Exploratory data analysis (EDA) was conducted to check for missing values and outliers and to inform input length selection for tokenization. EDA also revealed class imbalance, with 406 pairs labeled as "same," 144 as "exaggerates," and 113 as "downplays." Twenty percent of the data was held out as a test set, and the remaining data was partitioned into five folds for cross-validation, with model performance reported as the average across folds.

4.2 Problem Formulation

We formulate exaggeration detection as a supervised classification task over paired text inputs. Let $\mathcal{D} = \{(a_i, p_i, y_i)\}_{i=1}^{N}$ denote a dataset of N paired documents, where a_i is a scientific abstract, p_i is the corresponding press release, and $y_i \in \{0, 1, 2\}$ is the label indicating the relationship between the two texts. We define $y_i = 0$ as downplays, $y_i = 1$ as same, and $y_i = 2$ as exaggerates.

The model takes as input a concatenated representation of the abstract and press release of the form `[CLS] abstract_sentence [SEP] press_release_sentence [SEP]` and outputs a probability distribution over the three classes. The predicted label is the class with the highest probability.

4.3 Model Architectures

We evaluate three pretrained transformer encoder models, RoBERTa [8], PubMedBERT [4], and DeBERTa [5], along with a large language model (ChatGPT)[1]. RoBERTa is a general-domain language model trained on large-scale web and news corpora. In contrast, PubMedBERT, also referred to as BiomedBERT, is trained from scratch on biomedical text, enabling more specialized representations for scientific language. Importantly, RoBERTa and PubMedBERT share similar transformer encoder architectures, making them well-suited for isolating the impact of domain-specific pretraining. However, PubMedBERT is trained on less data than RoBERTa (107 GB vs. 160 GB), even when incorporating both PubMed abstracts and full-text articles, due to its restriction to domain-specific corpora. DeBERTa introduces architectural improvements through a disentangled attention mechanism and enhanced mask decoding, providing a strong general-domain baseline. In contrast to these encoder-based models, ChatGPT represents a large-scale generative language

model trained with instruction tuning, allowing us to evaluate whether modern large language models can match or exceed the performance of task-specific encoder models on scientific exaggeration detection.

4.4 Baseline Model and Treatments

First, to directly compare RoBERTa, PubMedBERT, and DeBERTa, we train baseline models in which only the classification head of each encoder is updated, while all pretrained encoder weights remain frozen. This serves as our control condition.

Given the limited size of our training data, we employ Low-Rank Adaptation (LoRA) as our primary treatment [6]. In this setting, pretrained model weights are frozen and trainable low-rank decomposition matrices are introduced into linear layers of the transformer architecture. This substantially reduces the number of trainable parameters, mitigating the chance of overfitting.

We additionally consider full-parameter fine-tuning as a secondary treatment, in which all model parameters are updated during training. This provides a comparison point to assess whether parameter-efficient adaptation can match or exceed the performance of standard fine-tuning.

We additionally evaluate a large language model (ChatGPT) using zero-shot and few-shot prompting. Unlike encoder-based models, ChatGPT is not fine-tuned; instead, we provide task-specific prompts with up to six labeled examples (two per class). This allows comparison under limited supervision.

4.5 Training Setup

We first conducted exploratory analysis to determine structural hyperparameters such as maximum sequence length, batch size, cross-validation strategy, and evaluation setup. We then tuned the learning rate, number of epochs, warmup ratio, and weight decay on RoBERTa, along with LoRA-specific parameters, to enable consistent comparison across models, following practices from prior work [11]. In subsequent iterations, we re-optimized the learning rate, weight decay, and warm up ratio for each model using Optuna. We identified all linear models as being optimal target modules for LoRA by experimentation, and we optimized LoRA hyperparameters, including rank (r), scaling factor (α), dropout with Optuna (Table S1).

All models were trained using five-fold cross-validation, with hyperparameters selected based on average performance across folds. Models were optimized using the fused AdamW optimizer in PyTorch with cross-entropy loss, and evaluated using macro-averaged F1. Early stopping with a patience of three epochs was applied to all models except those trained with LoRA.

4.6 Evaluation Metrics

Following hyperparameter selection, we conducted 30–50 experimental runs for each model using different random seeds to capture variability in model performance. For each run, models were trained on the training data and evaluated on the held-out test set, and the mean and standard deviation of the macro F1 score were computed across runs. We chose macro F1 scores as our primary metric due to class imbalance, as macro F1 scores equally weighs each class and emphasizes performance on underrepresented labels. The use of macro F1 also allows us to compare our results to previous publications in the field [13].

To evaluate whether domain-specific pretraining improves exaggeration detection, we compared macro F1 scores of the RoBERTa, PubMedBERT, and DeBERTa baseline models using the Friedman test, which is appropriate for comparing multiple models evaluated on the same data splits. Following a significant Friedman test, we performed a Nemenyi post hoc test to identify statistically significant pairwise differences between models. For comparisons between baseline models and their corresponding LoRA-trained variants, we applied the Wilcoxon signed-rank test to paired macro F1 scores. We also compared PubMedBERT and RoBERTa baselines with and without full parameter tuning using the same paired testing procedure. Statistical significance was assessed at $\alpha = 0.05$.

4.7 Quantization

The RoBERTa-base and PubMedBERT models each contain approximately 125M parameters. Although these parameter sizes are modest relative to modern large language models, we sought to assess whether model performance is preserved under post-training quantization, evaluating the feasibility of deploying exaggeration detection models in resource constrained environments.

We apply post-training quantization in two configurations: half-precision (FP16) and 8-bit integer (INT8). FP16 quantization reduces the bit-width of model parameters from 32 to 16 bits via dtype casting, preserving the floating-point representation while halving memory requirements. INT8 quantization requires a more deliberate mapping from the continuous floating-point space to a discrete integer grid. We use dynamic post-training quantization via `torch.ao.quantization.quantize_dynamic`, in which linear layer weights are statically quantized at quantization time using symmetric absmax INT8 quantization: the scale factor is derived from the maximum absolute weight value and the zero-point is fixed at zero. Activations are quantized dynamically at inference time; the scale and zero-point are computed from the actual activation distribution observed during each forward pass[3]. This approach contrasts with static post-training quantization, which requires a separate calibration pass over representative data to pre-compute activation statistics. We chose dynamic quantization as it requires no calibration data and is well-suited to classification tasks with variable input distributions.

5 Results and Discussion

To evaluate our hypothesis that domain-specific pretraining improves performance on exaggeration detection, we first conducted a Friedman test to assess whether there were overall differences in macro F1 scores across models on the held-out test dataset. The test indicated a statistically significant difference between models ($p < 0.001$). Given this result, we performed a Nemenyi post-hoc test to identify pairwise differences. The analysis showed that PubMedBERT significantly outperformed both DeBERTa ($p = 0.0013$) and RoBERTa ($p < 0.001$), while no statistically significant difference was observed between DeBERTa and RoBERTa (p = 0.34). These findings are consistent with the observed mean macro F1 scores, where PubMedBERT achieved the highest performance and DeBERTa and RoBERTa performed similarly (Table 1). All models outperformed a majority-class baseline (macro F1 = 0.252), with the exception of DeBERTa under full parameter fine-tuning, which collapsed to predicting the majority class across all test cases in each experiment.

Model	F1	SD
RoBERTa		
Frozen	0.289	0.031
LoRA	0.432	0.040
Full	0.419	0.041
PubMedBERT		
Frozen	0.330	0.038
LoRA	**0.453**	0.042
Full	0.434	0.049
DeBERTa		
Frozen	0.309	0.054
LoRA	0.299	0.038
Full	0.252	0.000

Table 1: Test macro F1 performance across models and fine-tuning strategies.

We observed class imbalance effects, with all models achieving higher precision and recall on the "same" class compared to "exaggerates" and "downplays." This is consistent with prior work on this dataset and suggests that minority classes are more challenging across models [13]. We additionally observed a consistent gap between cross-validation (CV) and held-out test performance across all models, with CV macro F1 scores exceeding test macro F1 scores. This gap may reflect optimistic bias introduced during Optuna hyperparameter tuning on CV folds, as well as the limited dataset size (n=530), which increases the risk of overfitting. Despite this, relative model performance remains consistent between CV and test evaluations (Table S2).

To evaluate post-training quantization (PTQ), we compared model performance and size under FP32, FP16, and INT8 precision (Table 2). FP16 reduced model size by approximately 50% with no loss in macro F1 for either model. INT8 quantization produced mixed results: PubMedBERT retained performance (+0.004), while RoBERTa showed a substantial decrease (-0.088). These findings suggest that PTQ can preserve per-

formance in some cases, particularly for domain-specific models, while providing significant reductions in model size.

Model	Precision	Macro F1	Δ F1	Size (MB)
RoBERTa	FP32	0.407	—	475.6
	FP16	0.407	0.000	237.8
	INT8	0.319	−0.088	230.9
PubMedBERT	FP32	0.462	—	417.7
	FP16	0.462	0.000	208.9
	INT8	0.466	+0.004	173.1

Table 2: Impact of post-training quantization on macro F1 and model size. ΔF1 is relative to the FP32.

To test our hypothesis that LoRA improves model performance in low-data settings, we applied the Wilcoxon signed-rank test to paired macro F1 scores comparing LoRA-trained models to their corresponding baselines. For both RoBERTa and PubMedBERT, LoRA significantly improved macro F1 scores on the held-out test dataset across runs ($p < 0.001$) (Table 1). In contrast, LoRA did not significantly improve the performance of DeBERTa relative to its frozen baseline ($p = 0.382$), suggesting that the benefits of parameter-efficient fine-tuning may depend on model architecture. We additionally performed full-parameter fine-tuning for RoBERTa and PubMedBERT and observed significant improvements in macro F1 scores on the held-out test dataset ($p < 0.001$), despite the small number of training examples. However, LoRA achieved similar or improved performance while updating a relatively small fraction of parameters (0.53% for RoBERTa, 2.39% for PubMedBERT; Table S3) compared to 100% for full-parameter fine-tuning, highlighting LoRA as a computationally efficient alternative. In contrast, full-parameter fine-tuning of DeBERTa significantly decreased performance ($p < 0.001$) and resulted in the model predicting only the majority class, as noted.

Zero-shot and few-shot prompting with ChatGPT 5.4 achieved macro F1 scores of 0.38 and 0.40, respectively, on the held-out test dataset. Despite this relatively strong performance under limited supervision, PubMedBERT with LoRA fine-tuning achieved higher macro F1 scores, indicating that task-specific adaptation of domain-specialized models remains advantageous over general-purpose large language models.

Overall, PubMedBERT with LoRA achieved the highest macro F1 score (0.453 ± 0.0417). While promising, this performance remains below the macro F1 score of 0.611 reported by Augenstein and Wright (2021) [13]. However, their approach leverages semi-supervised pattern-exploiting training (PET) and multi-task learning, which differs from our standard fine-tuning and LoRA-based approach. Our results demonstrate that domain-specific pretraining combined with parameter-efficient fine-tuning can achieve competitive performance without requiring more complex training pipelines.

6 Conclusion

Miscommunication of scientific findings to the general public remains a common and concerning issue. The goal of this study was to evaluate whether exaggeration in press releases could be detected relative to corresponding scientific abstracts using domain-specific models under limited data conditions. Our results show that the domain-specific pretrained model PubMedBERT outperformed both a general-domain model with a similar architecture (RoBERTa) and a more recent architecture with a larger training corpus (DeBERTa), as well as a large language model. LoRA further improved performance, achieving results comparable to or exceeding full fine-tuning while updating only a small fraction of parameters. Post-training quantization (PTQ) reduced model size while largely preserving performance for PubMedBERT. However, our study is limited by the small dataset and class imbalance, with the "same" category overrepresented relative to "exaggerates" and "downplays." Future work should address performance on underrepresented classes through techniques such as data augmentation, weighted loss functions, or cost-sensitive learning.

7 Author Attribution

Juba - Created code architecture that facilitates experiments across various models. This includes code pipeline for fine-tune methodology, quantization and hyper tuning. Report quantization and various sanity check and polishing.

Rebecca - Performed literature research on exaggeration detection in press releases. Performed Optuna optimization for frozen baseline models and LoRA models, ran experiments for RoBERTa, PubMedBERT, and DeBERTa classification only baseslines and LoRA optimized models. Carried out ChatGPT few shot and zero shot experiment. Performed statistical testing. Wrote draft of report except for sections on quantization.

8 Supplementary Information

Supplementary Table S1. Hyperparameter settings

Model	Hyperparameters
RoBERTa (Frozen)	lr = 0.0494, wd = 0.0688, warmup = 0.1957
PubMedBERT (Frozen)	lr = 0.0152, wd = 0.0690, warmup = 0.0543, epochs = 23
DeBERTa (Frozen)	lr = 0.0294, wd = 6.15e-05, warmup = 0.1915, epochs = 13
RoBERTa (LoRA)	lr = 4.97e-04, wd = 0.0456, warmup = 0.1202, r = 2, α = 16, dropout = 0.05
PubMedBERT (LoRA)	lr = 3.76e-04, wd = 0.0350, warmup = 0.1881, r = 16, α = 32, dropout = 0.10
DeBERTa (LoRA)	lr = 1.06e-05, batch = 8, wd = 0.0614, warmup = 0.1920, r = 2, α = 8, dropout = 0.10
RoBERTa (Full)	lr = 2.60e-05, wd = 0.0160, warmup = 0.0564
PubMedBERT (Full)	lr = 5.27e-05, wd = 0.3367, warmup = 0.0838

Supplementary Table S2. Comparison of model performance on CV and test data.

Model	Macro F1	Std Dev
RoBERTa		
Baseline (Test)	0.2888	0.0271
LoRA (Test)	0.4371	0.0435
Baseline (CV)	0.2756	0.0164
LoRA (CV)	0.4817	0.0353
PubMedBERT		
Baseline (Test)	0.3360	0.0371
LoRA (Test)	**0.4528**	0.0391
Baseline (CV)	0.4232	0.0154
LoRA (CV)	**0.5094**	0.0162

Supplementary Table S3. Training efficiency metrics by model and fine-tuning method. Peak GPU memory and total parameter updates are reported as medians across five cross-validation folds. Trainable% reflects the proportion of parameters updated during training.

Model	Method	Total Params	Trainable Params	Trainable%	Peak GPU (MB)	Param Updates
RoBERTa	Frozen	124.6M	0.6M	0.48%	**1,510**	188.5M
	LoRA	125.3M	0.7M	0.53%	1,833	**141.3M**
	Full	124.6M	124.6M	100%	5,380	66.1B
PubMedBERT	Frozen	109.5M	**2.3K**	<0.01%	1,604	**0.9M**
	LoRA	112.2M	2.7M	2.39%	2,186	994.7M
	Full	109.5M	109.5M	100%	4,865	46.4B

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